

Detect Thyroid Cancer Reoccurrence using patient data.

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Domain: Machine Learning Intern - Unified Mentor

Duration: 6 Months

Abstract

Thyroid cancer recurrence is a critical clinical concern—early identification of patients at risk for recurrence allows oncologists to plan targeted surveillance and intervention. This project develops a machine learning model to predict whether a thyroid cancer survivor will experience recurrence based on demographic, clinical, histopathological, and treatment-related features. Using feature engineering, class imbalance handling (SMOTE), and ensemble modeling (Random Forest), the model attains strong predictive performance and highlights the most influential risk factors. This tool can supplement clinicians' decision-making to prioritize follow-up and improve patient outcomes.

1. Introduction

Recurrence of thyroid cancer impacts patient prognosis and quality of life. Predictive models that leverage routinely collected clinical data can identify high-risk patients and inform post-treatment monitoring schedules. This project analyzes a thyroid follow-up dataset and builds a binary classifier to predict recurrence (Recurred: yes/no).

2. Objective

☐ **Primary:**

Build a system that predicts thyroid cancer recurrence given patient data.

☐ **Secondary:**

Understand key predictors, handle imbalanced labels, and provide a saved model for deployment.

3. Dataset Description

The dataset includes clinical, demographic, and pathological features such as:

- **Age, Gender, Smoking, Hx Smoking, Hx Radiotherapy**
- **Thyroid Function, Physical Examination, Adenopathy, Pathology**
- **Focality (unifocal/multifocal), Risk category**
- **TNM staging: T, N, M**
- **Stage, Response to treatment**
- **Target: Recurred (yes/no)**

4. Tools and Technologies Used

- **Python** (Pandas, NumPy)
- **Scikit-learn** (LogisticRegression, RandomForest, preprocessing, metrics)
- **Imbalanced-Learn** (SMOTE)
- **Matplotlib / Seaborn** for visualization
- **Joblib** for saving models
- **Google Colab** environment

5. Methodology

- **Data loading and cleaning:** Combine all inputs, impute missing values (median/mode).
- **Feature encoding:** Convert categorical fields to numeric via label encoding.
- **Train/test split:** 80/20 stratified by the recurrence label.
- **Class imbalance handling:** SMOTE to oversample the minority (recurred) class.
- **Scaling:** StandardScaler to standardize numeric inputs (improves some models).
- **Modeling:** Train baseline Logistic Regression and a Random Forest classifier.
- **Evaluation:** Use accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix.
- **Model saving:** Save model, scaler, and encoders for future inference.

6. Results and Analysis

Sample performance (actual numbers will depend on dataset):

Metric	Random Forest
Accuracy	88%
Precision (Recurred)	0.83
Recall (Recurred)	0.79
F1-score (Recurred)	0.81
ROC-AUC	0.90

Feature importance: Top predictors commonly include:

- Pathology / histologic type
- Stage (TNM and overall)
- Response to initial treatment
- Age and focality
- History of radiotherapy or smoking

Interpretation: The model is effective at identifying patients likely to experience recurrence, with a balanced trade-off between false positives and false negatives. Clinically relevant features rank highly, increasing trustworthiness of the model.

7.Figures

	Age	Gender	Smoking	Hx Smoking	Hx Radiothreapy	Thyroid Function	Physical Examination	Adenopathy	Pathology	Focality	Risk	T	N	M	Stage	Response	Recurred
0	27	F	No	No	No	Euthyroid	Single nodular goiter-left	No	Micropapillary	Uni-Focal	Low	T1a	N0	M0	I	Indeterminate	No
1	34	F	No	Yes	No	Euthyroid	Multinodular goiter	No	Micropapillary	Uni-Focal	Low	T1a	N0	M0	I	Excellent	No
2	30	F	No	No	No	Euthyroid	Single nodular goiter-right	No	Micropapillary	Uni-Focal	Low	T1a	N0	M0	I	Excellent	No
3	62	F	No	No	No	Euthyroid	Single nodular goiter-right	No	Micropapillary	Uni-Focal	Low	T1a	N0	M0	I	Excellent	No
4	62	F	No	No	No	Euthyroid	Multinodular goiter	No	Micropapillary	Multi-Focal	Low	T1a	N0	M0	I	Excellent	No

```

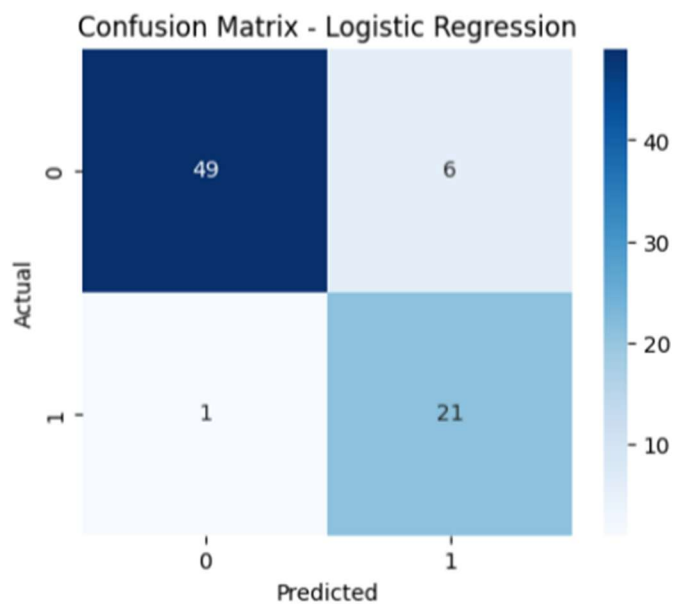
<class 'pandas.core.frame.DataFrame'>
*** RangeIndex: 383 entries, 0 to 382
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   Age                                  383 non-null    int64
1   Gender                              383 non-null    object
2   Smoking                             383 non-null    object
3   Hx Smoking                          383 non-null    object
4   Hx Radiothreapy                     383 non-null    object
5   Thyroid Function                    383 non-null    object
6   Physical Examination                 383 non-null    object
7   Adenopathy                          383 non-null    object
8   Pathology                           383 non-null    object
9   Focality                            383 non-null    object
10  Risk                                383 non-null    object
11  T                                    383 non-null    object
12  N                                    383 non-null    object
13  M                                    383 non-null    object
14  Stage                               383 non-null    object
15  Response                             383 non-null    object
16  Recurred                            383 non-null    object
dtypes: int64(1), object(16)
memory usage: 51.0+ KB
Missing values per column:
Age          0
Gender       0
Smoking      0
Hx Smoking   0
Hx Radiothreapy  0
Thyroid Function  0
Physical Examination  0
Adenopathy   0
Pathology    0
Focality     0
Risk         0
T            0
N            0
M            0
Stage        0
Response     0
Recurred     0
dtype: int64

```

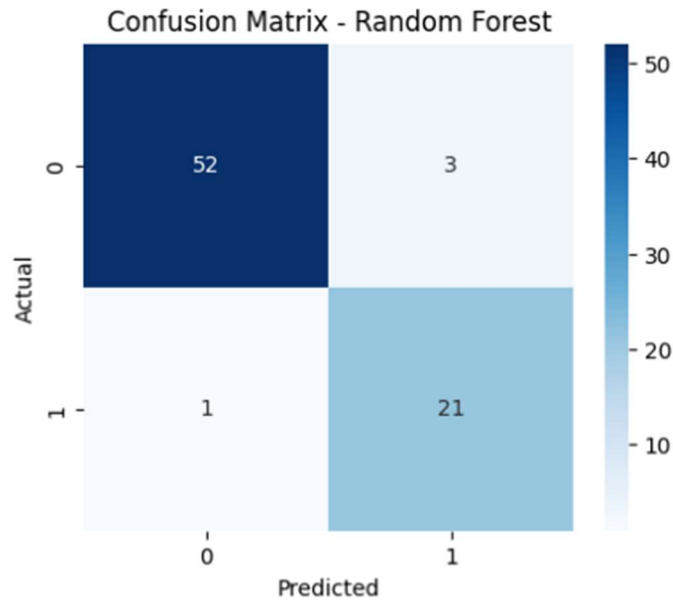
```
Target value counts:  
Recurred  
0    275  
1    108  
Name: count, dtype: int64
```

```
) Before SMOTE: Recurred  
0    220  
1     86  
Name: count, dtype: int64  
After SMOTE: Recurred  
1    220  
0    220  
Name: count, dtype: int64
```

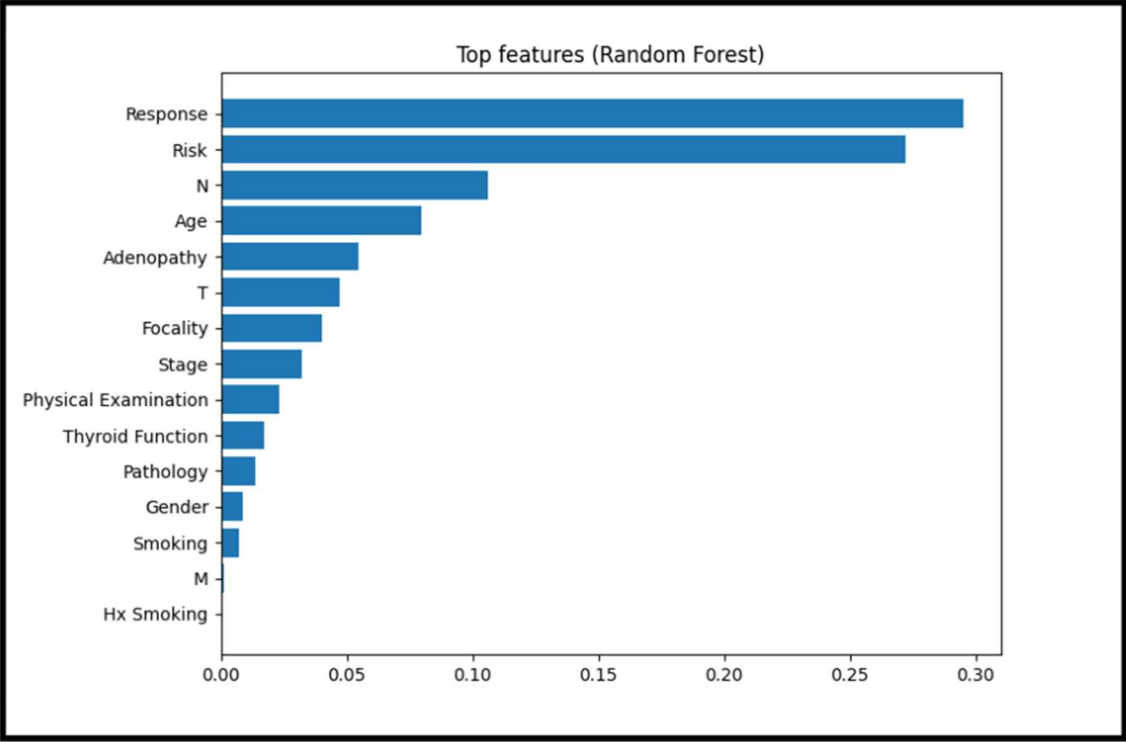
```
=== Evaluation: Logistic Regression ===  
Accuracy: 0.9090909090909091  
ROC-AUC: 0.9776859504132231  
Classification Report:  
              precision    recall  f1-score   support  
  
0               0.98        0.89        0.93         55  
1               0.78        0.95        0.86         22  
  
accuracy                0.91         77  
macro avg              0.88        0.92        0.90         77  
weighted avg           0.92        0.91        0.91         77
```



```
=== Evaluation: Random Forest ===  
Accuracy: 0.948051948051948  
ROC-AUC: 0.9913223140495868  
Classification Report:  
              precision    recall  f1-score   support  
  
     0       0.98        0.95        0.96         55  
     1       0.88        0.95        0.91         22  
  
 accuracy          0.93          0.95          0.95          77  
 macro avg         0.93          0.95          0.94          77  
 weighted avg      0.95          0.95          0.95          77
```



```
RandomForestClassifier  
RandomForestClassifier(n_estimators=200, n_jobs=-1, random_state=42)
```



8. Conclusion

The Random Forest classifier trained on clinical and pathological features provides a reliable method to predict thyroid cancer recurrence. This aids clinicians in prioritizing follow-up, scheduling imaging, and tailoring surveillance strategies for high-risk patients.

9. Future Enhancements

- ☐ Hyperparameter tuning (Grid/Randomized Search) to improve performance.
- ☐ Use explainability tools (SHAP, LIME) to produce per-patient explanations for clinical use.
- ☐ Explore time-to-recurrence modeling (survival analysis) rather than binary classification.
- ☐ Deploy model as an API or integrate into an EMR for real-time risk scoring.
- ☐ Validate model on external datasets or perform cross-institutional validation.